Verdant
FOODS

Verdant Foods Digital Platform · Phase 1

Product Architecture & UX Planning

Client Application (Farmers & VSLAs)

An Android app that works offline, for Vendor Onboarding & Verification, Batch Inventory & Tracking and the Digital VSLA Ledger.



Version 1.0 · October 2026
Reference: INV-NKL-00705
Based on the Verdant Foods Business Plan and Product Requirements (June 2026)

Prepared by NEXKOOMSON

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Planning for the Verdant Foods Client Application for farmers and VSLAs, written in plain language.

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The app in one page

One app for the people who grow the food.

Verdant's document calls it the Client Application. In Phase 1 it serves farmers and savings groups (VSLAs).

Farm

Farm profile, harvest logging with trace numbers, stock and orders

Savings group

Members, group rules, meetings, loans and the ledger

Money

Payments due and paid, to MoMo or bank

Works offline

Saves on the phone first, sends when signal returns

Built with	Flutter, for Android first
Size	Under 15MB, so it installs on low-cost phones, as Verdant's document requires
Languages at launch	English, Twi, Dagbani and Dagaare, with voice prompts
Shoppers	Buy on the AgTech Marketplace website in Phase 1, not in this app

Who uses the app

Four kinds of app user

Farmer

Uses a phone, often with weak signal. Needs few steps and clear words.

VSLA member

Sees their own savings, shares and loans.

VSLA admin

Records meetings, members and group rules for the whole group.

Extension officer

Signs up farmers and groups in the field and sends farms for checking.

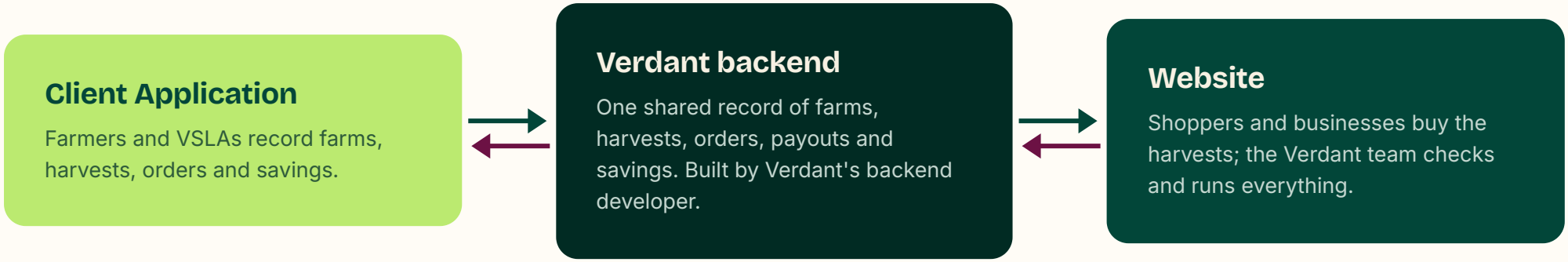
User	Farm space	Log harvest	Money	Savings group	Group admin tools	Sign up others
Farmer	●	●	●	◐	–	–
VSLA member	–	–	◐	◐	–	–
VSLA admin	–	–	◐	●	●	–
Extension officer	◐	◐	–	◐	–	●

● Full use ◐ Own records, or helping someone else – No access

One person can be a farmer and a VSLA member at the same time; the app shows both spaces.

How the app fits the platform

Same data, different door.



A harvest logged in the app	appears on the AgTech Marketplace and in the B2B Procurement Engine once approved
An order placed on the website	appears in the farmer's Orders tab
A farm sent for checking	appears in the Admin Interface queue
A payment to a farmer	shows in the Money tab, and can go into their savings group

How the app is organised

Two spaces, one simple app.

Farm space

Home
Today's orders, collections, money due

Log harvest
The big central button

Stock
Each harvest, what's left, price

Orders
Accept and hand over

Money
Payments due and paid

My farm
Location, size, methods, check status

Savings group space

Overview
Group savings, loans out, cash box

Meeting
Record everyone in one pass

Members
Add members, see balances

Loans
Give out, repay, interest

Ledger
Every transaction, in order

Group rules
Share price, interest, cycle

Sign in with phone number and PIN. A switch at the top moves between the two spaces. Language and sync status are always one tap away.

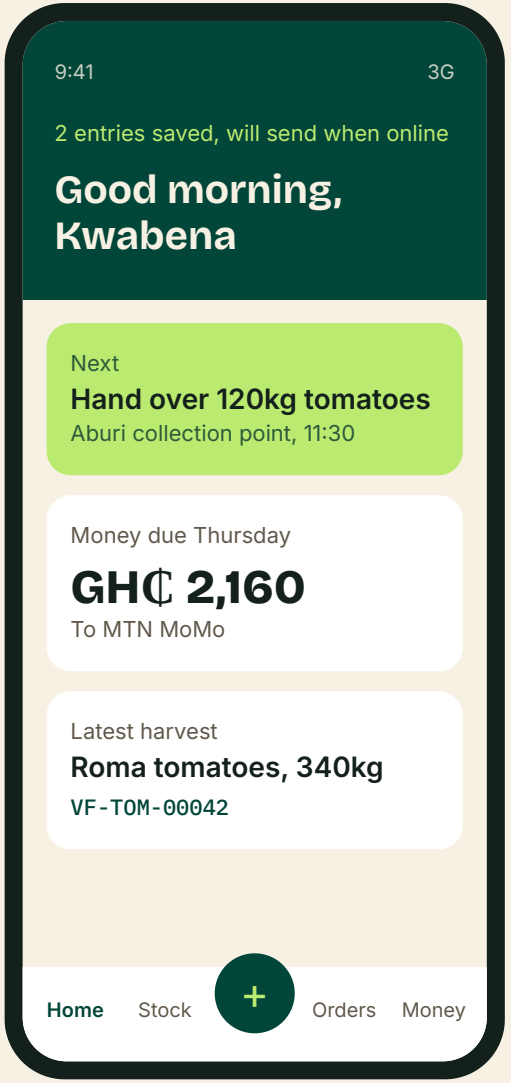
Farmer screens

One thumb, weak signal, one clear next step.

An Android app that works offline and stays under 15MB. Everything saves on the phone first and sends when signal returns. It is available in English, Twi, Dagbani and Dagaare from launch.

Tab	What the farmer does
Home	Sees today's orders, collections and money due
Log harvest	Records a harvest in five steps and gets its trace number at once
Stock	Sees each harvest, what is left and its price
Orders	Accepts orders and hands them over at collection
Money	Payments due and paid to MoMo or bank

Farm profile and the savings group sit under the farmer's name at the top.



Logging a harvest

Five steps, under two minutes, with or without signal.

1

Crop

Pick from pictures: tomato, garden egg, yam, maize

2

How much

Big number pad, then kg, crate or bunch

3

When and where

Harvest date and which plot

4

Photo and methods

One photo; tick the farming methods used

5

Price and save

Set price, tap save

Right after saving

VF-TOM-00042

The trace number appears at once, even offline, from a reserve of numbers kept on the phone.

What this covers in Verdant's document

Batch Inventory & Tracking: every harvest is a batch with a permanent trace number. Photos and farming methods are added to the farm's record. The photos are always real, taken by the farmer.

Savings groups (VSLA)

The group's paper ledger, now safe on a phone.

Members, group rules, weekly savings, loans and the cash box, recorded in one pass at the meeting, even with no signal.

Overview

Group savings, members, loans out, cash in the box, next meeting

Meeting

Record each member's shares and repayments in one go

Members

Add members, see each person's balance

Loans

Amount, interest rate, schedule, what's been repaid

Ledger

Every transaction in order. Nothing can be quietly changed.

Group rules

Share price, loan interest rate, cycle length

Phase 1 starts with 20 savings groups

Savings and payouts move by MTN MoMo, Telecel Cash or AT Money. Balances are shown large and clear, with the time they were last updated.

Comes later

Credit scores, crop insurance and small investments, from Phase 3.

Key journeys

Farmer

1	2	3	4	5	6	7	8
Sign up	**Farm checked**	**Add farm**	**Log harvest**	**Set price**	**Get order**	**Hand over**	**Get paid**
Phone and PIN, with help	Methods approved	Location and size	Gets a trace number	Goes on sale	Accept or decline	At collection point	To MoMo or bank

Goal: log a harvest in under two minutes, with or without signal.

Savings group meeting

1	2	3	4	5	6
Set up group	**Open meeting**	**Record savings**	**Loans**	**Close meeting**	**Ledger updated**
Members and rules	Works offline	Each member in one pass	Give out or repay	Group agrees the totals	Cash box balance saved

Goal: as fast as the paper book, and the totals match the cash counted.

Working offline

No signal is normal. The app never waits for it.

1

Save on the phone

Every entry is stored on the phone first

2

Show it's waiting

"2 entries saved, will send when online"

3

Send when signal returns

Works on 2G, 3G or 4G, in the background

4

Confirm

"All sent", with the time

Situation	What the app does
Logging a harvest offline	Saves it and gives the final trace number straight away
Recording a savings meeting offline	Saves the whole meeting; totals are locked once the group agrees
The same record changed twice	The newest change wins; the VSLA admin is told if a meeting record clashes
Money figures	Always show when they were last updated, so nobody trusts an old balance

Verdant's document asks for this offline-first mode. The backend developer builds the receiving side of the sync.

Languages and payments

Speak the farmer's language. Pay the way farmers get paid.

Languages in the app

Language	Where it's spoken most	Available
English	Nationwide	At launch
Twi	Ashanti, Eastern and more	At launch
Dagbani	Northern Region	At launch
Dagaare	Upper West Region	At launch
Gonja	Northern and Savannah	Pilot year
Waale	Upper West Region	Pilot year
Ewe	Volta Region	Pilot year
Ga	Greater Accra	Pilot year

Translated by native speakers. Key steps also have voice prompts.

Where money goes

MTN MoMo

Telecel Cash

AT Money

Ghanaian banks

Farmers	Paid to MoMo or a bank account
Savings groups	Deposits and payouts by MoMo

Look and feel

Verdant owns green. The food brings the colour.

The two greens come from the logo. The bright colours come from the references we chose: lively like a good market, always anchored by Verdant green.

Verdant green #024639	**Night green** #012B23	**Leaf** #5F9F0D

Fresh lime #BBEA70	**Orange** #FF7006	**Tomato** #E5412B	**Sun** #F7C33C	**Blush** #F4C4CF	**Plum** #6C1144

Cream #F7F1E3	**Stone** #ECE5D6	**Grey text** #62594C	**Ink** #13201B

Website and shop use the full range for categories, offers and illustrations.

Business and team tools stay calm: mostly cream and green, with colour only in charts and status.

Farmer app is green and lime. **Plum** marks the savings group space.

Look and feel

Built for thumbs, sunlight and small screens

Big buttons

+ Log harvest

Accept order

At least 48 pixels tall. Labels in plain words with an icon.

Sync banner

2 entries saved, will send when online

All sent · 9:41

Always visible at the top of the Farm and Savings spaces.

Money card

Group savings

GH₵ 18,450

Updated today, 9:41

Big figures, small decimals, and the time last updated.

Strong contrast for outdoor light, text never below 14px on the phone, and pictures next to crop names.

Testing and release

Three ways for Verdant to try the app.

1

Browser preview

A link on Vercel to click through the screens on any phone or laptop. Best for design and flow reviews.

2

Test installs

Testers install from an email invite (Firebase App Distribution, free) and get every update. The real test of offline use.

3

Play Store

Internal testing, then public release, through Verdant's own Google Play account.

Test on a low-cost phone	A phone like the ones farmers use, to check speed and size
Test with weak signal	Switch data off and on during harvest logging and meetings
Test in every launch language	Native speakers check the wording and voice prompts

Roadmap

Start simple, add finance later.

Months 1 to 6 | Months 7 to 18 | Month 19 onward

Phase 1: pilot

20 savings groups

- Farm profile and checks
- Harvest logging with trace numbers
- Stock, orders and money
- Savings groups: rules, meetings, loans, ledger
- Offline saving and sync
- English, Twi, Dagbani, Dagaare

Phase 2: tracing and delivery

- Collection and delivery steps in the app
- Printing QR crate labels at collection
- Gonja, Waale, Ewe and Ga

Phase 3: finance

- Weather-based crop insurance
- Carbon rewards for farmers
- Credit scores from savings and sales
- Small investments

Who builds what

Clear lines, so connecting the backend needs no redesign

	NEXKOOMSON builds	Verdant's backend developer builds	We agree together
Design	All app screens in Figma, in every state		Status names and messages
App	The Flutter app, test builds and Play Store release		
Sign-in	Phone and PIN screens, who sees what	Accounts and permissions	The list of roles
Offline	Saving on the phone and sending when online	Receiving and merging offline entries	How phones and server sync
Tracing	Harvest logging and trace number display	Trace numbers and the reserve given to phones	Number format
Money	Payout and savings screens	MoMo and bank connections	Payment status names
Languages	App wording in every language, voice playback		Translations by native speakers, at cost

What we need from the backend

Eight things that make connecting smooth

1 **Technical documentation** of what the backend offers

2 **Phone and PIN sign-in** and roles

3 **The fields** for farms, harvests, orders, groups, members and loans

4 **How offline entries are synced**

5 **How trace numbers are reserved** for phones

6 **Photo upload rules**

7 **Payout status names** for MoMo and bank

8 **Test logins and test addresses**

Already set by Verdant's document: secure connections, encrypted data and identity checks for wallets. We build around these.

How we work and what's next

Design with the website, build after approval.

1
Plan
Done: this document

2
App screens in Figma
Farm space, Savings space, offline states

3
Figma prototype
Clickable, for Verdant to try

4
Approval
Sign off the design

5
Build
Flutter app, Android first

6
Test installs
Farmers and VSLAs try it on real phones

7
Connect backend
Sync, payouts, trace numbers

8
Play Store release
Through Verdant's account

Estimated build: 6 to 8 weeks after the design is approved.

Decisions made

Key decisions for the app

Platform	Flutter, Android first, under 15MB, works offline
Who it's for	Farmers and VSLAs. Shoppers use the website in Phase 1.
Extension officers	Yes, from Phase 1
Trace numbers	Made by the backend, with a reserve kept on each phone
Payouts	MTN MoMo, Telecel Cash, AT Money or a Ghanaian bank account
Languages	English, Twi, Dagbani, Dagaare at launch; Gonja, Waale, Ewe, Ga in the pilot year
Photos	Real photos taken by farmers on every harvest record
Play Store	Released under Verdant's own account (US$25 one-time)

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Based on the Verdant Foods Business Plan and Product Requirements (June 2026), the Verdant logo and the reference images chosen by NEXKOOMSON. Where this plan adds ideas beyond Verdant's document, they are shown as our recommendations.

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